Reionisation as Preconditioned Transparency in Darke Theory Timing, Source Burden, and Topology Beyond Source-Only Percolation

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March 2026

Abstract

The standard account of cosmic reionisation treats the process as the progressive ionisation of a largely neutral intergalactic medium by the first luminous sources, with morphology determined primarily by source clustering, escape fractions, and recombination sinks. This paper develops a stricter Darke-theoretic alternative. The central claim is not that sources are irrelevant, but that radiative transparency may have been preconditioned by uneven early structural commitment. In this view, reionisation is not merely a photon-budget problem. It is the large-scale unveiling of a universe whose geometric legibility had already become non-uniform.

The argument is organised as a three-front test programme. First, the timing sector is reconstructed under current observational constraints from CMB optical depth and late-stage Lyα evidence, showing that the standard window is disciplined enough that source uncertainty remains load-bearing. Secondly, the source sector is re-examined in light of JWST-era compact early populations, including closure-node candidates such as Little Red Dots, to test whether ordinary galaxy-only budgets remain sufficient without hidden tuning. Thirdly, the topology sector is developed as the decisive discriminator: if reionisation was purely source-driven, ionised morphology should reduce to sources plus sinks; if transparency was preconditioned, coherent residual structure should remain after conditioning on the source field.

A concrete technical test plan is then stated. The plan defines standard and Darke-style hypotheses, identifies measurable topology signatures, specifies residual tests against source-field conditioning, and outlines a structured confrontation with present and near-future datasets. The goal is not to overclaim completed resolution, but to establish a disciplined whitepaper framework for testing whether the epoch of reionisation records only astrophysical ignition, or also deeper uneven closure in the early cosmological substrate.

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1 Introduction

The epoch of reionisation is often described in one sentence: after recombination the universe became mostly neutral, the first luminous sources switched on, ionised bubbles grew, those bubbles merged, and the intergalactic medium became transparent again. That sentence is useful, but it compresses too much. It hides three separate questions that must not be conflated:

1. **Timing**: when reionisation began, how rapidly it progressed, and by what redshift it was substantially complete;
2. **Sources**: which physical populations supplied the ionising photons and whether those populations close the budget without severe tuning;
3. **Topology**: what the geometry of ionised and neutral regions looked like as the transition unfolded.

Within ordinary cosmological language these questions are related but still basically source-centred. Reionisation is treated as the radiative consequence of early structure formation. This paper does not deny that luminous sources matter. It challenges a deeper assumption: that the geometry of transparency was determined only by where those sources happened to form.

The Darke-theoretic alternative developed here is stronger. It proposes that reionisation may have been preconditioned. The earliest transparent regions may not simply have been those with the largest ionising emissivity. They may also have been those in which the universe had already become more structurally committed, more closure-bearing, and therefore more radiatively legible. If so, the reionisation record is not merely a source census written in hydrogen. It is a map of uneven early cosmological unveiling.

The present whitepaper has five aims:

1. to state the standard timing, source, and topology burdens with sufficient technical discipline;
2. to formulate a Darke-style alternative in which transparency can be structurally preconditioned;
3. to identify exact observational signatures that distinguish source-only percolation from preconditioned transparency;

4. to define a residual-testing programme after source-field conditioning;
5. to provide a full technical test plan suitable for later data-driven execution.

This paper is therefore not a finished empirical claim. It is a framework paper in the precise sense. Its purpose is to define the load-bearing structure of the investigation before numerical execution and confrontation with data. The aim is not to declare that reionisation has already been structurally re-explained. The aim is to define the exact point at which the standard picture would fail, and the exact point at which a preconditioned-transparency alternative would begin to earn explanatory necessity.

2 Foundational Setting in Darke Theory

The present construction stands downstream of the core Darke-theoretic lineage

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure.}$$

That sequence is not repeated here in full formal detail; it is already carried by the broader corpus. What matters for reionisation is the following compressed interpretation.

2.1 Closure and admissibility

Darke Theory distinguishes between a regime being formally describable and a regime being genuinely supportable. Closure is the threshold at which recursively generated distinction becomes self-integrating persistence. Admissibility is the further filter determining whether that persistence can actually be carried across the burden required by the regime.

Reionisation then becomes more than an ionisation history. It becomes a transition in legibility. A region is not merely bright or dim. It is more or less available to sustained radiative transparency depending on the degree to which the relevant geometric and structural commitments have already stabilised.

2.2 Preconditioned transparency

The central proposal of this paper may be stated in one sentence:

Reionisation was not only the ionisation of a neutral medium by early emitters; it may have been the radiative exposure of a universe whose transparency field had already become unevenly structured.

This is not a denial of sources. It is a claim about background readiness. In standard language, the medium is passive except for density, sinks, and radiative transfer. In the present framework, the medium may carry a prior structural field influencing the ease with which ionisation propagates and persists.

2.3 Relation to the existing corpus

Several parts of the broader Darke corpus already touch the present problem:

- the inflation-replacement paper supplies a non-primitive early-universe threshold logic;

- the *What Big Bang?* special rejects the Big Bang as deepest ontology and places early cosmological legibility downstream of more primitive structural emergence;
- the *Little Red Dots* paper places unusually compact high-redshift systems directly in the key redshift range relevant to the reionisation transition;
- the black-hole papers support the possibility of early compact accretors and closure-dominant source classes.

The present work does not merely quote those results. It asks whether reionisation provides a common observational arena in which they may interact.

2.4 A refinement-side contextual equation

In the wider Darke framework, early cosmological emergence is associated with refinement under finite closure capacity. A representative form already used elsewhere in the corpus is

$$\frac{dR}{dt} = \mu(R - R_c)\, f\left(\frac{R}{R_{\max}}\right),$$

where R denotes an effective refinement rank, R_c a symmetric critical state, and $R_{\max}$ the closure capacity. The present paper does not assume that reionisation is itself described directly by this equation. Rather, the point is structural: if the early universe approaches geometric commitment through thresholded refinement, then later radiative transparency need not emerge into a perfectly uniform background of readiness. The reionisation-era medium may inherit structured non-uniformity from earlier closure history.

3 Notation Summary

For convenience, the principal symbols used in the present paper are collected here.

Symbol	Meaning
R	Effective refinement rank in the wider early-universe contextual equation.
R_c	Symmetric critical state in the refinement-side framework.
$R_{\max}$	Closure capacity in the refinement-side framework.
τ	Thomson optical depth to reionisation.
σ_T	Thomson cross section.
$n_e(z)$	Free-electron density history.
$f_{\rm esc}$	Escape fraction of ionising photons from galaxies.
$\xi_{\rm ion}$	Ionising photon production efficiency.
$\rho_{\rm UV}$	Non-ionising ultraviolet luminosity density.
$\dot{n}_{\gamma,\rm AGN}$	Escaping ionising emissivity from accreting systems.
$S(x)$	Local source field.
$K(x)$	Local sink or recombination field.
$\rho(x)$	Local ordinary density environment.
$C(x)$	Effective structural-commitment field used to parameterise preconditioned transparency.
$h(C)$	Monotone function encoding structural preconditioning.

$P_{\rm ion}(x)$	Local ionisation-propagation or transparent-persistence probability in schematic form.
$B(x)$	Bubble-size or ionisation proxy in the source-conditioned confrontation.
$R_B(x)$	Residual ionisation-topology field after source conditioning.

4 What Standard Cosmology Must Explain

The standard model does not merely need reionisation to happen somehow. It must satisfy three distinct burdens.

4.1 Timing burden

Reionisation must satisfy the integrated Thomson optical depth inferred from CMB polarisation while also remaining compatible with late-stage Lyα and quasar constraints. In modern form this means the following:

- the optical depth τ must remain near the low Planck-era value, disfavouring excessively early or overly extended reionisation histories;
- the end stages must be largely completed by approximately $z \sim 5.3$–6;
- the patchy phase cannot be stretched arbitrarily without tension from kinetic Sunyaev–Zel'dovich and related constraints.

A convenient integrated observable is

$$\tau = c\,\sigma_T \int_0^{z_{\rm max}} n_e(z)\,\frac{dt}{dz}\,dz,$$

where σ_T is the Thomson cross section and $n_e(z)$ the free-electron density history. This does not by itself determine morphology, but it constrains how early and how long the transition may run.

Thus the standard timing window is no longer arbitrarily flexible. It is not sufficient to state that reionisation happened "sometime early." The allowed history is already disciplined by integrated optical-depth limits and by the observationally accessible tail of the transition. More strongly: the timing window is bounded tightly enough that source efficiency and topology remain consequential. Reionisation is not free to be both arbitrarily late and arbitrarily weak, nor arbitrarily early and arbitrarily extended.

Representative benchmark values are now sufficient to make the timing burden concrete. Planck-era analyses favour a low optical depth of order $\tau \approx 0.054$, while modern late-tail studies have explicitly explored scenarios in which reionisation remains ongoing until approximately $z \approx 5.2$–5.5. The standard timing problem is therefore not merely qualitative. It must fit a low integrated optical depth while still allowing enough late neutral structure to satisfy the observational tail without overextending the patchy phase.

4.2 Source burden

Standard cosmology must also close the ionising photon budget. The core unknowns are well known:

- the abundance of faint galaxies below direct detection,
- the ionising photon production efficiency $\xi_{\rm ion}$,
- the escape fraction $f_{\rm esc}$,
- the role of faint or obscured AGN,
- the effective sink structure of the IGM.

A minimal budget identity may be written schematically as

$$\dot{n}_{\gamma,\rm esc}(z) = f_{\rm esc}(z)\,\xi_{\rm ion}(z)\,\rho_{\rm UV}(z) + \dot{n}_{\gamma,\rm AGN}(z),$$

where $\rho_{\rm UV}$ is the non-ionising ultraviolet luminosity density and $\dot{n}_{\gamma,\rm AGN}$ the escaping ionising contribution from accreting systems.

The critical point is that the standard source budget is often stated more confidently than it is actually closed. Even when galaxy-dominated solutions are viable, viability often depends upon assumptions that remain poorly constrained. The source problem is not merely one of naming a plausible source class in principle. It is one of whether the standard galaxy-only closure remains robust without hidden tuning.

Published JWST-era analyses now permit a non-negligible contribution from accreting systems to the ionising budget, so the source burden can no longer be treated as a trivially galaxy-only problem even when galaxies remain the default dominant class.

4.3 Topology burden

Finally, standard cosmology must explain reionisation morphology. Ionised bubbles, their merger history, neutral-island persistence, and the large-scale connectivity of the transition must all emerge from source clustering, sink physics, and ordinary density structure. This burden is often underemphasised, but it is where the standard and Darke-style pictures can be most sharply separated.

5 Reionisation as a Three-Front Problem

The present paper therefore treats reionisation as a three-front investigation.

5.1 Front I: timing

The first question is whether the standard timing envelope is broad and forgiving or narrow and load-bearing. If the envelope is very broad, source and topology tests weaken. If it is narrow, source uncertainty matters sharply.

5.2 Front II: sources

The second question is whether standard source populations close the budget comfortably. If they do, compact early populations are less decisive. If they do not, then closure-node systems, compact red populations, and early black-hole seeds become immediately relevant.

5.3 Front III: topology

The third question is the decisive one. Even if timing is acceptable and source budgets are plausible, the ionised morphology may still contain coherent residual structure after conditioning on the source field. If so, source-only percolation is incomplete.

6 Standard Topological Expectations

To build a discriminator we must state the standard expectations precisely enough to be falsified.

Definition 6.1 (Standard source-driven reionisation topology)**.** *A reionisation morphology is source-driven in the standard sense if, after conditioning on source positions, source luminosity, escape fractions, sink physics, and large-scale density environment, the ionised and neutral topology is statistically explainable without invoking any additional organising field.*

Remark 6.2. *This standard topology picture must be represented fairly. Modern reionisation modelling does not reduce morphology to a naive cartoon of isolated spherical bubbles. It already includes nontrivial percolation behaviour, late neutral-island persistence, and increasingly sophisticated topological diagnostics. The present paper therefore does not oppose a primitive standard picture. It asks whether even these improved source-and-sink descriptions are sufficient once residual structure is tested after conditioning.*

In operational form, standard source-driven topology implies the following.

6.1 Bubble size should track source field

Larger ionised regions are expected preferentially around brighter and more clustered sources. The bubble-size distribution should evolve as a source-weighted percolation problem modified by mean free path and sinks.

6.2 Late neutral islands are residuals, not structural lags

Large neutral islands may survive into the late stages, but they are expected to arise from source scarcity, self-shielding, and sink competition rather than from any deeper unevenness in the transparency substrate itself.

6.3 Residual morphology should collapse under source conditioning

Once the source field is included, any remaining coherent topological residue should be small enough to be interpreted as noise, finite-volume scatter, or model imperfection rather than as evidence for another organising principle.

7 Darke-Style Alternative: Preconditioned Transparency

We now state the competing hypothesis.

Definition 7.1 (Preconditioned transparency)**.** *Let $C(x)$ denote an effective structural-commitment field on the reionisation-era cosmic medium. Reionisation is said to be preconditioned if the local ease of ionisation and transparent persistence depends not only on source emissivity and sinks, but also on $C(x)$, with higher $C(x)$ corresponding to greater effective radiative legibility.*

This field is not introduced here as a completed microphysical operator. It is an effective organising variable used to test whether the morphology demands more than source percolation alone.

7.1 Minimal modelling form

A minimal local propagation law may be written schematically as

$$P_{\text{ion}}(x) = g\big(S(x), K(x), \rho(x)\big)\, h\big(C(x)\big),$$

where:

- $S(x)$ is the local source field,
- $K(x)$ is the sink/recombination field,
- $\rho(x)$ is the ordinary density environment,
- $h(C)$ is a monotone function encoding structural preconditioning.

If $h \equiv 1$, the model reduces to the standard source-driven case. If h varies nontrivially and improves the explanation of observed topology after standard conditioning, the preconditioned picture gains support.

Operationally, $C(x)$ is not intended here as a completed microphysical field. It is an effective structural variable whose role is to bias one or more of the following: local transparency readiness, propagation ease of ionisation fronts, persistence of ionised connectivity, or resistance of a region to reverting into late neutral-island structure.

Proposition 7.2 (Standard-case reduction)**.** *If $h(C) \equiv 1$ throughout the reionisation-era medium, then the preconditioned-transparency propagation law reduces identically to the standard source-driven propagation form.*

Proof. Substituting $h(C) \equiv 1$ into

$$P_{\text{ion}}(x) = g\big(S(x), K(x), \rho(x)\big)\, h\big(C(x)\big)$$

yields

$$P_{\text{ion}}(x) = g\big(S(x), K(x), \rho(x)\big).$$

Thus all dependence on the structural field $C(x)$ disappears, and the propagation law becomes entirely source-, sink-, and environment-conditioned in the standard sense.

7.2 Physical interpretation

The point is not that photons no longer matter. The point is that the universe may not have been equally ready to be revealed everywhere. Reionisation bubbles would then trace both luminosity and prior structural readiness.

8 Topology Signature List

The standard and Darke-style pictures can now be separated by an explicit signature list.

8.1 Standard signature set

Under the standard picture, one expects:

1. bubble size to correlate strongly with source brightness and clustering;
2. topology to evolve mainly by source-centred percolation and sink-limited island survival;
3. morphology to become largely explainable once sources and environment are conditioned on;
4. no large-scale directional organisation beyond ordinary structure and survey effects.

8.2 Darke-style signature set

Under the preconditioned-transparency picture, one instead expects:

1. bubble topology too coherent for the source field alone;
2. characteristic scales that remain unexpectedly stable after source conditioning;
3. galaxy–bubble mismatch, with some regions too ionised or too neutral relative to observed source content;
4. late neutral islands that are too structured for sink-limited residuals alone;
5. residual large-angle directional memory;
6. disproportionately strong association of compact early populations with transparency topology.

9 A Full Technical Test Plan

The paper now turns from conceptual framing to a concrete investigation architecture.

9.1 Hypotheses

H0: Reionisation morphology is fully explained by sources, sinks, and ordinary environment.

H1: After conditioning on the standard variables, reionisation retains coherent topological residuals consistent with preconditioned transparency.

9.2 Phase I: timing anchor

Objective. Lock the allowed reionisation timing window so that later tests are meaningfully constrained.

Tasks.

a) collect CMB optical-depth constraints and late-stage Lyα/quasar constraints;

b) define an onset floor, midpoint window, and completion ceiling;

c) assess whether ordinary source-budget models operate comfortably inside that envelope or only with tuned assumptions.

Deliverable. A timing envelope table with allowed early, middle, and late reionisation windows.

Interpretation. If the timing envelope is tight, source and topology tests become load-bearing. If the envelope is too broad, structural interpretation weakens.

9.3 Phase II: source-budget closure test

Objective. Determine whether galaxies alone close the ionising budget or whether compact early populations reduce hidden tuning.

Tasks.

a) compute baseline galaxy-only emissivity under a grid of $f_{\rm esc}$, $\xi_{\rm ion}$, and faint-end assumptions;

b) add candidate compact populations, including LRD-like systems and AGN-like classes;

c) quantify how much each additional population relaxes parameter tension.

Deliverable. A source-budget matrix with rows for galaxy-only, galaxy+compact, and galaxy+compact+AGN scenarios.

Interpretation. If ordinary galaxies close the budget only under narrow assumptions while compact populations robustly relax that pressure, the source sector begins to favour the Darke-style picture.

9.4 Phase III: bubble–source residual test

Objective. Test whether inferred ionised environments track the source field cleanly.

Tasks.

a) infer local bubble or ionisation proxies around source ensembles;

b) fit a predictive relation of the form

$$B(x) = F\big(L(x), n_{\rm src}(x), A_{\rm AGN}(x), \rho(x), K(x)\big),$$

where B is a bubble-size or ionisation proxy;

c) map and analyse the residual field

$$R_B(x) := B_{\mathrm{obs}}(x) - B_{\mathrm{pred}}(x).$$

Deliverable. Residual maps, residual distributions, and coherence diagnostics on R_B.

Interpretation. If R_B becomes noise after source conditioning, the standard picture survives. If R_B retains coherent spatial organisation, source-only percolation is incomplete.

9.5 Operational criterion for residual topology detection

A positive residual-topology detection is not claimed from arbitrary mismatch between data and a fitted source model. The residual field must satisfy at least one of the following conditions after source, sink, and ordinary environment conditioning:

1. statistically significant spatial autocorrelation beyond that expected from controlled null ensembles;
2. persistent topological excess relative to standard mocks, as measured by one or more of Betti numbers, persistent-homology summaries, Minkowski functionals, or skeleton connectivity measures;
3. large-scale residual organisation that remains stable under rotated-mask, random-axis, or environment-matched control tests;
4. systematic source–bubble mismatch that is structured rather than noise-like across the observational field.

A residual claim therefore requires organised survival of morphology after conditioning, not merely imperfect fit quality.

Proposition 9.1 (Residual falsifier of source-only sufficiency)**.** *Suppose a source-conditioned predictive model has already included source positions, source luminosities, sink structure, and ordinary environment. If the residual topology field satisfies one or more of the operational detection criteria above in statistically controlled confrontation with standard mocks, then source-only sufficiency is falsified on the tested observational class.*

Proof. Under source-only sufficiency, all coherent explanatory content relevant to topology is exhausted by the conditioned variables and by the null ensemble built from the standard model class. Therefore any post-conditioning residual should be reducible to noise, finite-volume scatter, or ordinary model imperfection. If instead the residual field survives in one of the operationally defined coherent forms above after controlled comparison with standard mocks, then the tested standard class has failed to absorb the observed morphology. Source-only sufficiency is therefore falsified on that tested class.

9.6 Phase IV: topology comparison

Objective. Determine whether the observed morphology resembles ordinary percolation or a more constrained unveiling process.

Metrics.

- bubble-size distribution,
- island-size distribution,
- percolation threshold behaviour,
- Euler characteristic / genus,
- Minkowski functionals,
- persistent homology / Betti numbers,
- skeleton and connectivity measures.

Tasks.

a) evaluate the metrics on standard mock fields;

b) evaluate the same metrics on observationally inferred fields;

c) evaluate the same metrics on Darke-style structured mocks including $C(x)$.

Deliverable. A multi-metric topology confrontation panel.

Interpretation. If the data consistently show excess connectivity, preferred scales, or over-persistent islands relative to the standard mocks, topology begins to favour preconditioning.

9.7 Phase V: late neutral island test

Objective. Use the tail of reionisation as the cleanest stress point.

Tasks.

a) identify late-island candidates from Lyα and quasar-tomography style constraints;

b) measure island size, persistence, symmetry, clustering, and environmental relation;

c) compare with sink-limited predictions.

Deliverable. A late-island anomaly scorecard.

Interpretation. If late islands are too structured, too coherent, or too persistent for sink-limited residuals alone, the Darke-style picture gains a particularly strong foothold.

9.8 Phase VI: directionality test

Objective. Test whether topology carries large-angle directional memory after source and mask control.

Tasks.

a) construct hemisphere and rotated-mask comparisons for topology metrics;

b) build random-axis and mask-matched control ensembles;

c) test whether one or more topology measures remain unusually axis-sensitive after control.

Deliverable. A directionality significance panel.

Interpretation. Any robust large-angle residual organisation after control would be difficult to absorb into source-only percolation.

9.9 Phase VII: Darke-style structured mock field

Objective. Provide an explicit alternative rather than only a critique.

Definition 9.2 (Structured transparency mock)**.** *A structured transparency mock is a simulation field in which ionisation probability depends on an effective structural field $C(x)$ in addition to the ordinary source and sink terms.*

Tasks.

a) construct a toy field $C(x)$ with tunable coherence length and amplitude;

b) define ionisation propagation through $P_{\text{ion}}(x)$ as above;

c) compare the resulting topology against both standard mocks and data.

Deliverable. A minimal Darke-style morphology simulator.

Interpretation. The aim is not yet proof. It is to show whether the qualitative data lean more naturally toward preconditioned transparency than toward source-only percolation.

10 Decision Framework

At the end of the full programme, the result may be classified into one of four bins.

A. Standard sufficient:
timing, source budget, and topology are all adequately explained by the standard picture.

B. Standard sufficient but tuned:
standard cosmology survives, but only under narrow source assumptions or parameter choices.

C. Source anomaly only:
timing and topology are acceptable, but the standard source budget remains under strain without compact early populations.

D. Structural anomaly:
after source and environment conditioning, topology retains coherent residual structure not well reproduced by standard percolation mocks.

The fourth class is the decisive Darke-style target.

11 Work Packages

For practical execution, the investigation should be organised into explicit work packages.

WP1: Timing envelope

- Collect CMB τ and late Lyα constraints.
- Build a timing window summary table.
- Identify the part of parameter space already under strain.

WP2: Source census

- Build source-budget scenarios.
- Insert compact high-z populations and early AGN candidates.
- Quantify whether compact populations reduce tuning.

WP3: Bubble–source confrontation

- Gather bubble-size or ionisation proxies.
- Fit the standard predictive model.
- Build residual maps.

WP4: Topology diagnostics

- Implement bubble/island statistics.
- Add topology descriptors and skeleton measures.
- Compare observations against standard and structured mocks.

WP5: Late neutral islands

- Compile island constraints.
- Test size, persistence, and structure.
- Score standard versus Darke-style explanations.

WP6: Directionality and control tests

- Run rotated-mask and random-axis nulls.
- Test for residual directional organisation.

WP7: Structured mock simulator

- Implement the $C(x)$-weighted propagation model.
- Generate comparison suites.
- Identify which observables are most discriminating.

12 Datasets and Observational Access

The present programme naturally divides into what can be done now and what requires future data.

12.1 What can be done now

- CMB optical-depth timing constraints from Planck-era low-ℓ polarisation analyses;
- Lyα and quasar-based late-stage reionisation constraints;
- JWST high-redshift source populations, including compact red source classes and candidate early accretors;
- published bubble-inference frameworks using ensembles of galaxies and Lyα spectroscopy in the JWST era;
- published or near-published neutral-island recovery methods using photometric intergalactic-medium tomography.

12.2 What will strengthen later

- 21-cm tomography of ionised bubbles and neutral islands,
- direct bubble morphology against source catalogues,
- large-area topology and directionality tests.

13 Main Technical Claim of the Whitepaper

The entire investigation can be compressed to one exact question:

After conditioning on source populations, sinks, and ordinary environment, does reionisation still contain coherent topological residuals that are too organised for source-driven percolation alone?

That is the blade of the whole programme.

If the answer is no, the Darke-style interpretation is weakened. If the answer is yes, then reionisation ceases to be only an astrophysical ignition problem and becomes evidence for uneven early transparency in the cosmological substrate.

14 Conclusion

This whitepaper has built a full technical framework for testing whether the epoch of reionisation was merely source-driven or was also structurally preconditioned. The argument was organised in three fronts: timing, sources, and topology. Timing constrains the allowed history sharply enough that source uncertainty remains load-bearing. Sources remain unsettled enough that compact high-redshift populations cannot yet be dismissed as peripheral. Topology is the decisive discriminator, because it can retain coherent residual structure after source conditioning.

The paper has therefore not asserted completed victory for the Darke-style picture. It has done the more important prior work: it has defined a discipline. It has stated what standard cosmology must explain, what a preconditioned-transparency alternative would have to look like, how the two can be separated observationally, and how the investigation should be executed in concrete phases.

The decisive issue is no longer whether reionisation occurred. It is whether its geometry reduces cleanly to source-centred percolation once sources, sinks, and ordinary environment are conditioned on. If it does, the standard picture survives this confrontation. If it does not, and coherent topology remains, then reionisation becomes evidence not merely of early astrophysical ignition but of uneven prior transparency in the cosmological substrate itself.

The strongest falsifier of the present framework is correspondingly plain. If a sufficiently rich source-plus-sink-plus-environment model removes the residual topology under the operational criteria stated in this paper, then preconditioned transparency is not needed on that tested class. If no such standard model class succeeds, then the burden shifts. Reionisation must then be treated not only as the ignition of early luminous sources, but as a record of deeper uneven readiness in the cosmological medium itself.

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